

Renu Shekhawat

L-11, Department of Mathematics, Indian Institute of Science, Bengaluru, Karnataka - 560 012
renushekhawa@iisc.ac.in

Employment

Indian Institute of Science

Research Associate
Department of Mathematics
Mentor : [Prof. Tirthankar Bhattacharya](#)

Bengaluru, India
Feb 2026 - Present

Education

Indian Statistical Institute, Bangalore Centre

Philosophiae Doctorate in Mathematics
Supervisor : [Prof. Soumyashant Nayak](#)

Bengaluru, India
Jul 2019 - Feb 2026

University of Rajasthan

Master of Mathematics

Jaipur, India
Aug 2016 - Aug 2018

Sophia Girls' College

Bachelor of Mathematics

Ajmer, India
Jul 2013 - Jul 2016

Research Interests

Functional analysis, Operator algebras

Publications / Preprints

- S. Nayak, R. Shekhawat; *On the Convergence of the Normalized Power Sequence of Spectral Operators on Hilbert Space* (2024), available at : <https://arxiv.org/abs/2410.16318>
(Accepted for publication in JOT)
- S. Nayak, R. Shekhawat; *On the Jordan-Chevalley-Dunford decomposition of operators in type I Murray-von Neumann algebras* (2025), available at : <https://arxiv.org/abs/2506.17227>
(Submitted for publication).
- S. Nayak, R. Shekhawat; *On the Convergence of the Normalized Power Sequence of Riesz Operators* (2026), available at : <https://arxiv.org/abs/2606.13743>
(Submitted for publication).

Thesis

On the Jordan-Chevalley-Dunford Decomposition of Certain Classes of Operators and Convergence of Their Normalized Power Sequences. (Submitted to Indian Statistical Institute in June 2025).

Fellowships / Scholarships

- **Junior Research Fellowship**

(Declined due to acceptance of the Institute fellowship at ISI)

- University Grants Commission, Government of India. Dec 2018
- Council of Scientific and Industrial Research, Department of Science and Technology, Government of India. Jun 2017

- **Inspire Scholarship for Higher Education**

Department of Science and Technology, Government of India. Jul 2013 - Jul 2018

Teaching Assistant / Grader

- Real Analysis, B.math(I) - Semester I Aug 2025 - Nov 2025
Instructor : Prof. B. V. Rajarama Bhat
- Function spaces, B.math(III) - Semester I Aug 2023 - Nov 2023
Instructor : Prof. Soumyashant Nayak
- Complex Analysis, M.Math(I) - Semester II Jan 2022 - May 2022
Instructor : Prof. Ramesh Sreekantan
- Complex Analysis, B.Math(III) - Semester I Sep 2021 - Dec 2021
Instructor : Prof. Mathew Joseph

Conferences / Workshops Attended

- **Ergodic Group Actions and Unitary Representations III** Warsaw, Poland
Institute of Mathematics of the Polish Academy of Sciences, Warsaw Feb 2 - 6, 2026
- **School on Invariant Structures in Group Actions** Będlewo, Poland
Research and Conference Centre of the Institute of Mathematics of the Polish Academy of Sciences, Będlewo Jan 11 - 16, 2026
- **Young Mathematicians in C^* Algebras (YMC * A) 2025** Odense, Denmark
University of Southern Denmark (SDU), Odense Jul 21 - 25, 2025
- **International Workshop on Operator Theory and its Applications (IWOTA) 2025** Enschede, Netherlands
University of Twente, Enschede Jul 14 - 18, 2025
- **20th Workshop: Noncommutative Probability, Operator Algebras and Related Topics, with Applications** Będlewo, Poland
Research and Conference Centre of the Institute of Mathematics of the Polish Academy of Sciences, Będlewo Jul 6 - 12, 2025
- **Young Mathematicians in Operator Algebras** Delhi, India
Indian Statistical Institute, Delhi Centre Mar 24 - 28, 2025

- **Conference on Operator Algebra and Related Topics (COART)** Mumbai, India
Indian Institute of Technology, Bombay Feb 24 – 29, 2025
 - **Noncommutative Mathematics and Applications (NCMA)** Bengaluru, India
Indian Statistical Institute, Bangalore Centre Oct 24 – 26, 2024
 - **Young Mathematicians in Operator Algebras** Delhi, India
Indian Statistical Institute, Delhi Centre Feb 26 – Mar 02, 2024
 - **International Conference on Spectral and Approximation Theory (ICSAT)** Kochi, India
Cochin University of Science and Technology, Kerala Nov 27 – 30, 2023
 - **Conference on Functional Analysis and Related Topics (CFART)** Mumbai, India
Indian Institute of Technology, Bombay Feb 21 – 25, 2023
 - **IWM (Indian Women and Mathematics) Annual Conference** Pune, India
Indian Institute of Science Education and Research, Pune Dec 27 - 29, 2022
 - **Indo-French Centre for Applied Mathematics (IFCAM)** Bengaluru, India
Summer School on Mathematical Aspects of Quantum Mechanics
Indian Institute of Science, Bengaluru Jun 01 – 12, 2022
-

Talks

- **Invited:**
 - For a minicourse, as an *early Career Researcher* in **Simons Semester “Invariant Structures in Groups, Spaces, and Algebras”**
Institute of Mathematics of the Polish Academy of Sciences, Warsaw Jan 11 - Feb 15, 2026
 - For a talk, *A spatial form of Yamamoto’s theorem for spectral operators*,
Analysis and Probability Research Group (APRG) Seminar,
Indian Institute of Science, Bengaluru Sep 22, 2025
- **Contributed Talks :**
 - *A stronger form of Yamamoto’s theorem on singular values*
IWOTA 2025, University of Twente, Enschede Jul 18, 2025
 - *Jordan-Chevalley-Dunford decomposition in Type I Murray-von Neumann algebras*
20th Workshop: Noncommutative Probability, Operator Algebras and Related Topics,
with Applications, Będlewo Jul 10, 2025
 - *Convergence of the normalized power sequence for spectral operators*
COART, Indian Institute of Technology, Bombay Feb 27, 2025
 - *On Matrices over $C(X)$ for X a Stonean space*
ICSAT, Cochin University of Science and Technology, Kerala Nov 27, 2023
- **Seminar Talks :**

- *An elementary computational approach to the Jordan-Chevalley decomposition and the Jordan canonical form.*

PDF - RS Annual Symposium, Indian Statistical Institute, Bangalore Centre

Jan 31, 2025

- *Convergence of the normalized power sequence for spectral operators*

QP - Day at ISI, Indian Statistical Institute, Bangalore Centre

Jan 15, 2025